

Merge Sort

Definition

Merge Sort is a **divide-and-conquer** sorting algorithm. It divides the array into smaller sub-arrays, sorts them, and then **merges** the sorted sub-arrays to produce the final sorted array.

Algorithm

1. Divide the array into two halves.
2. Recursively divide each half until each sub-array has one element.
3. Compare elements of the two sub-arrays.
4. Merge them in sorted order.
5. Continue merging until the complete array is sorted.

Output:

```
Enter number of elements: 5
Enter element: 7
Enter element: 9
Enter element: 3
Enter element: 4
Enter element: 6
Original array: [7, 9, 3, 4, 6]
Sorted array: [3, 4, 6, 7, 9]
Time Complexity:
Best Case:  $O(n \log n)$ 
Average Case:  $O(n \log n)$ 
Worst Case:  $O(n \log n)$ 
Space Complexity:  $O(n)$ 

...Program finished with exit code 0
Press ENTER to exit console. 
```

Time Complexity

Case	Complexity
Best Case	$O(n \log n)$
Average Case	$O(n \log n)$
Worst Case	$O(n \log n)$

Space Complexity

- **Auxiliary Space:** $O(n)$
- Extra space is required for merging the sub-arrays.